

If you are studying physics in high school then refer *Physics* 5th edition by Halliday, Resnick, Krane. This guide is intended for those who want to complete a core education in physics. I am assuming you have mathematical knowledge up until mentioned in this [PDF](#). We start with introductory physics and then work our way through the undergraduate core as well as electives you may take.

1. Mathematics that you will encounter throughout your undergraduate degree.

- *Mathematical Methods in the Physical Sciences* by Boas
- *Mathematical Methods for Physicists* by Arfken, Weber, Harris

2. We go through basic mechanics, electrostatics and magnetostatics and modern physics.

- *Fundamental Laws of Mechanics* and *Basic Laws of Electromagnetism* by Irodov
- *Introduction to Mechanics* by Kleppner, Kolenkow and *Introduction to Classical Mechanics* by Morin
- *Electricity and Magnetism* by Purcell, Morin [supplement with *Div, Grad, Curl, and All That* by Schey]
- *Vibrations and Waves* by French and *Waves* by Crawford
- *Introduction to Quantum Mechanics* by Griffiths, Schroeter and *A Modern Approach to Quantum Mechanics* by Townsend

3. We go through the undergraduate core of a physics degree.

- *Classical Mechanics* by Taylor
- *Introduction to Electrodynamics* by Griffiths [supplement with *Notes on Vector Calculus* by Tong]
- *Thermal Physics* by Schroeder or *Fundamentals of Statistical and Thermal Physics* by Reif
- *Optics* by Hecht
- *Principles of Quantum Mechanics* by Shankar

4. Mathematics that you will encounter throughout your graduate years and beyond.

- *Mathematical Physics: A Modern Introduction to Its Foundations* by Hassani
- *Mathematical Methods of Classical Mechanics* by Arnold
- *Advanced Mathematical Methods for Scientists and Engineers* by Bender, Orszag

5. The graduate core of a physics PHD.

- *Modern Classical Mechanics* by Helliwell, Sahakian and *Classical Mechanics* by Goldstein, Poole
- *Modern Electrodynamics* by Zangwill and *Classical Electrodynamics* by Jackson
- *Statistical Physics* by Kardar
- *Quantum Mechanics Lecture Notes* by Littlejohn and *Lecture Notes on Topics in Quantum Mechanics* by Tong

Try to find lecture notes on all topics you are studying of appropriate level such as [David Tong's Lecture Notes](#) and equivalent. Advice on undergraduate electives as well as graduate electives in physics can be found here in this [post](#). One can only learn physics by solving problems so solve a lot and try to understand. For PHD qualifying / graduate qualifying, exams these texts are quite good to go through.

- *Problems in General Physics* by Irodov
- *Problems and Solutions on Mechanics, Electromagnetism, Quantum Mechanics, Thermodynamics and Statistical Mechanics, Optics, Atomic/Nuclear/Particle Physics, Solid State Physics, and Relativity and Cosmology* by Lim Yung-Kuo
- *Princeton Problems in Physics with Solutions* by Newbury, Newman, Ruhl, Staggs, Thorsett
- *A Guide to Physics Problems* by Cahn, Nadgorny
- The online qualifying exam archives of Princeton, Berkeley, Stanford, MIT, Chicago and Harvard, which are free and have the real exams.

A few books that have really good problems are listed below:

- *Analytical Dynamics of Particles and Rigid Bodies* by Whittaker
- *Static and Dynamic Electricity* by Smythe
- *Problems in Classical Electromagnetism* by Cherny, Kuz'mich / *Problems in Electrodynamics* by Batygin, Toptygin
- *Collection of Problems in Classical Mechanics* by Kotkin, Serbo
- *Problems in Mathematical Physics* by Budak, Samarskii, Tikhonov
- *Exploring Quantum Mechanics* by Galitski, Karnakov, Kogan, Galitski.
- *Selected Problems in Quantum Mechanics* by ter Haar (from Gol'dman, Krivchenkov).
- *Statistical Mechanics: An Advanced Course with Problems and Solutions* by Kubo.